

Photoconducting Polymers

Introduction :-

Photoconductive polymers absorb electromagnetic radiation and produce an increase of electrical conductivity.

Electrical conductivity is usually very small in organic compounds. Conducting polymers usually have large electrical conductivity.

Photoconductive polymer is a smart material based on conductive polymer, and the electrical conductivity can be controlled by the amount of radiation.

Photoconductivity is an optical and electrical phenomenon, which material's electrical conductivity increase by absorption of electromagnetic radiation (eg. visible light, ultraviolet light, infrared light).

Photoconductive polymers can serve as good insulators when the electricity, free electrons and holes are absent.

In general, the polymers usually satisfy these two features.

1. Photoconductive polymers can absorb light to excite electrons from ground state to excited state. The photoexcited electron will form a pair of charge carriers, it can be separated by electric field.

2. Photoconductive polymers must allow migration of either photoexcited electrons or holes or both through the polymer in the electric field towards the appropriate electrodes.

Photoconductive polymers act merely as charge transporting media, and it can be p-type or n-type, however most known photoconductive polymers are p-type (only transport holes).

The main advantage of organic photoconductive polymers are -

- ① near-IR sensitivity
- ② panchromaticity - sensitive to all visible colours of the spectrum.
- ③ flexibility for application
- ④ simple fabrication
- ⑤ low cost.

Currently, the best organic photoconductive polymers are as sensitive as the inorganic devices based on selenium.

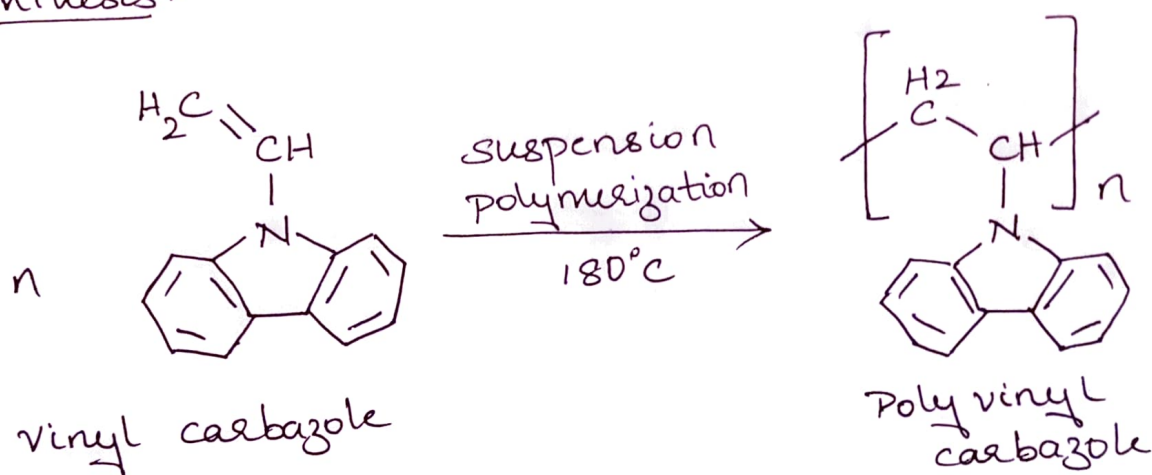
Poly vinyl carbazole

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Synthesis:-

Polyvinylcarbazole (PVC) is a temperature-resistant thermoplastic polymer produced by radical polymerization from the monomer N-vinyl carbazole. It is photoconductive and thus the basis for photorefractive polymers and organic light emitting diodes.

Synthesis:-



Polyvinyl carbazole is obtained from N-vinyl carbazole by radical polymerization in various ways. It can be produced by suspension polymerization at 180°C with sodium chloride and potassium chromate as catalyst.

Properties:-

Physical

1. It can be used at temp upto 160-170°C & is therefore a temperature resistant thermoplastic.

2. The electrical conductivity changes depending on the illumination. For this reason, PVK is classified as a semiconductor or photoconductor.
3. It is extremely brittle, but the brittleness can be reduced by copolymerization with a little isoprene.

chemical

1. It is soluble in aromatic hydrocarbons, halogenated hydrocarbons and ketones.
2. It is resistant to acids, alkalis, polar solvents and aliphatic hydrocarbons.
3. The addition of polyvinylcarbazole to other plastic masses increases their temperature resistance.

Application :-

It is used in insulation technology, electrophotography (copiers and laser printers), organic light emitting diodes and photovoltaic devices.

Laser printers:

Laser printing is an electrostatic digital printing process. It produces high quality text and graphics by repeatedly passing a laser beam back and forth over a negatively charged

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cylinder called a 'drum' to get a charged image. The drum can selectively collect electrically charged powdered ink (toner) and transfers the image to paper.

As digital photocopiers, laser printers employ a xerographic printing process.

Xerography ~~is~~ electrophotography is a photocopying technique. Xerography use photoconductive polymer as a foundation material, which is a solid chemical.

However, laser printing differs from analog photocopiers. Because the image is produced by direct scanning of the medium across the printer's photoreceptor, which enables laser printing to copy images more quickly than most photocopiers.

The first laser printer was invented by Xerox PARC in the 1970's.